NETFLOW

SOLID MATERIAL LEVEL CONTROL · NTF-POD2 SERIES

ROTARY PADDLE LEVEL SWITCH

Rotary paddle level sensor for full / empty detection in silos, bunkers and containers

The NTF-POD2 series rotary paddle level switch mechanically detects the level of powder and granular solids in silos. The paddle, driven at 5 rpm by a geared synchronous motor, is stopped when covered by material; the resulting reaction movement is detected by a limit switch, a relay contact output is generated and the motor stops.

IP66 / IP67 PROTECTION CLASS	**3 Steps** SENSITIVITY SETTING
24 VDC / 220 VAC SUPPLY OPTIONS	**5 rpm** PADDLE SPEED



NTF-POD2 · L TYPE PADDLE

- 100 / 200 mm shaft standard, 200 / 500 / 1000 mm extension shaft optional
- Limit switch / micro switch detection
- SS316 stainless steel shaft and paddle
- 1-1/2" BSPP connection, 1" and 2" optional
- L and T type paddle, selected per application
- Glass-fibre reinforced PC-ABS connection; aluminium and stainless optional

TYPICAL USE

1. **VERTICAL MOUNTING · HIGH LEVEL** Mounted from the silo roof. Used for full-level alarm and filling stop.
2. **SIDE MOUNTING · INTERMEDIATE LEVEL** Mounted horizontally on the silo wall. For high or intermediate level control.
3. **ANGLED MOUNTING · LOW LEVEL** Mounted at an angle on the cone. Used for empty alarm and feed demand.

APPLICATION AREAS

CEMENT · READY-MIX	GYPSUM · LIME · AGGREGATE	FEED · SEED · FLOUR	PLASTICS · COAL DUST

OPERATING PRINCIPLE

1
EMPTY STATE

The geared synchronous motor rotates the paddle shaft at 5 rpm. As long as the shaft turns freely, the relay stays in the empty state and the LED is green.

2
MATERIAL CONTACT

When material covers the paddle, a counter force acts on it. This force rises until it exceeds the threshold set by the 3-step spring adjustment.

3
SIGNAL OUTPUT

The load is detected by the limit switch; the motor stops and the changeover contact switches within max. 0.6 s, the LED turns red. When material recedes, the system resets.

ELECTRICAL DATA

Supply (24 VDC)	24 VDC ±3%, max. 2.7 W
Supply (220 VAC)	220 VAC, max. 3.1 W
Signal output	1 changeover contact (C / NO / NC), 24 VDC / 220 VAC 3 A
Signal delay	Max. 0.6 s
Terminal	Max. 2 mm² (AWG 14) cable
Electrical connection	PG 9 cable gland · optional DIN 43650 or M12 4-pin
Status indicator	LED — empty: green · full: red
Protection class	IP66 / IP67

OPERATING CONDITIONS

Ambient temperature	-20 °C … +80 °C
Process temperature	-5 °C … +80 °C · +160 °C with cooling extension
Silo pressure	-0.5 … +0.5 bar
Max. vibration	5–500 Hz, 3G RMS (IEC 60068-2-64)

MECHANICAL DATA

Housing	Option P: PC-ABS · Option A: machined aluminium
Drive	Geared synchronous motor
Detection	Limit switch (micro switch)
Shaft bearing	Double ball bearing
Sealing	NBR seal (Viton optional)
Connection material	Glass-fibre reinforced PC-ABS · aluminium · stainless optional
Connection size	1-1/2" BSPP (standard) · 1" and 2" BSPP optional
Shaft and paddle	Stainless steel SS316
Shaft length	100 mm (standard) · 200 mm · optional extension shaft
Shaft speed	5 rpm

Sensitivity setting	3-step spring
Min. bulk density	250 g/l (with standard L paddle)
Weight	870 g (100 mm shaft) · 1000 g (200 mm shaft)

PADDLE TYPES

L TYPE

Standard. Powders and fine granules, density above 250 g/l.

T TYPE

Large surface. Earlier detection in light and fluffy materials.

ELECTRICAL CONNECTION

220 VAC MODEL

N (1)	Neutral
L (2)	Phase
COM (3)	Common
NC (4)	N. Closed
NO (5)	N. Open

24 VDC MODEL

NC (1)	N. Closed
NO (2)	N. Open
COM (3)	Common
− (4)	0 V
+ (5)	24 VDC

SIGNAL STATUS

SILO EMPTY	C—NC closed · LED green · paddle rotates
SILO FULL	C—NO closed · LED red · motor stops

CONNECTION TYPES

PG 9 cable gland	Standard
DIN 43650 (ABS) connector	Optional
M12 4-pin connector	Optional

MECHANICAL INSTALLATION

01 The unit must be mounted perpendicular to the surface, away from the filling stream.

02 The paddle must not touch the silo wall, and material must not jam between them.

03 In horizontal mounting, the cable gland must face down to prevent water ingress.

04 In low-level applications, a guard must be fitted above the paddle.

05 After installation, tighten the housing cover and gland fully.

06 Do not mount on vibrating cones or vibrating surfaces.

TROUBLESHOOTING

SYMPTOM	POSSIBLE CAUSE	REMEDY
Shaft not rotating, LED off	No supply or wrong wiring	Measure the supply voltage at terminals.
Full signal with empty silo	Material jammed between paddle and wall	Clean the paddle, increase distance or add guard.
No signal with material	Density below 250 g/l or spring step too high	Lower the spring step; change to T type paddle.
Moisture in housing	Gland facing up or cover not closed	Turn gland down, check cover and seal.
Signal flickering	Mounted on vibrating surface	Move away from vibrating cone.

SELECTION GUIDE

HOUSING: A plastic housing is sufficient for indoor and dusty applications. For outdoor or impact-risk lines, aluminium is preferred.

SHAFT LENGTH AND PADDLE: For sticky materials, choose 200 mm shaft or extension. For light materials below 250 g/l, use T type paddle.

HOT PROCESSES: Where process temperature exceeds +80 °C, use cooling extension model (up to +160 °C).

OPTIONS

- Aluminium housing (machined)
- Cooling extension model, +160 °C process
- 200 / 500 / 1000 mm extension shaft
- T type paddle
- 1" and 2" BSP connection
- Aluminium or stainless connection
- DIN 43650 (ABS) or M12 4-pin connector
- Viton seal

ISO 9001 · EN 61000-6-4 · EN 61000-6-2 · EN 61010-1

Contact our technical team to select the right shaft length, paddle type and connection size for your application. NET FLOW · netflw.com · info@netflw.com · +90 553 296 52 42